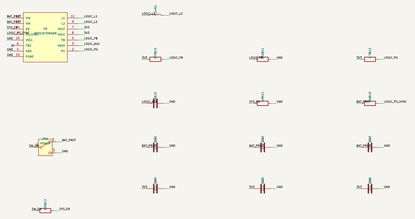
01 / 3.3 V BUCK-BOOST — 500 mA minimum design target



Draft exact inductor suffix, capacitor MPNs and loop layout still require verification.

02 / BLE CONTROLLER + UART PROGRAMMING



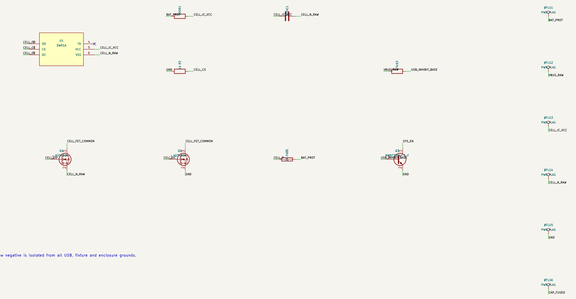
GPIO2 high for boot, then TRIG_BOJ output LOW before ARM. GPIO7 reads TRIGGER_ALLOWED.

03 / 2.5 F EFFECTIVE CAP BANK — regulated 4.8 V



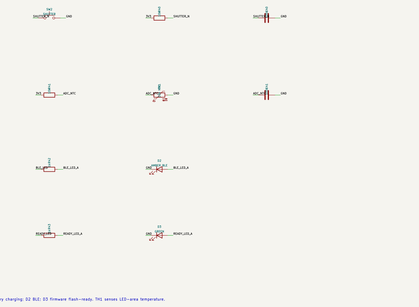
Two series cells: midpoint connects to LTC3225 CX. No bleed balancing resistors.

09 / CELL FUSE + DW01 LOW-SIDE PROTECTION



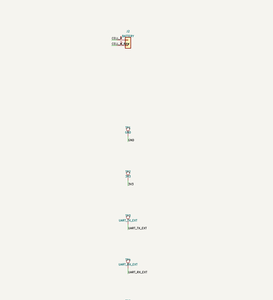
Raw negative is isolated from all USB, fixture and enclosure grounds.

04 / SHUTTER, INDICATORS, LED-AREA NTC

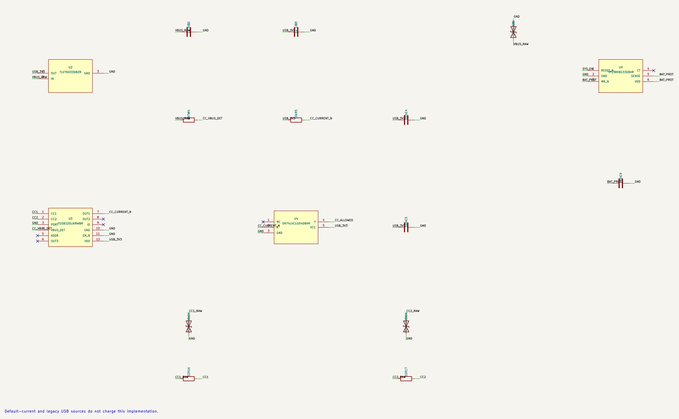


D1 indicates battery charging. D2 BLE: D3 firmware flash-ready. TH1 senses LED-area temperature.

05 / CONNECTORS + 3.3 V UART POGO ACCESS

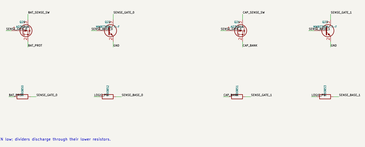


10 / TYPE-C SOURCE QUALIFICATION — CHARGING ONLY



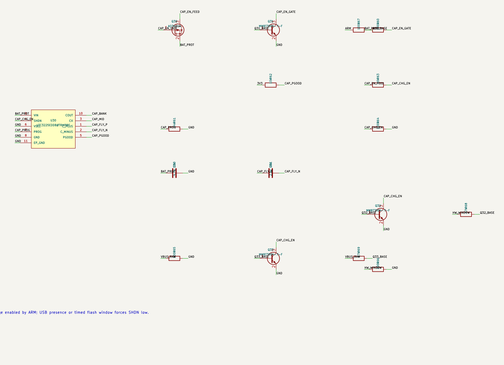
Default-current and legacy USB sources do not charge this implementation.

07 / ADC POWER-OFF ISOLATION — enabled by regulator PG



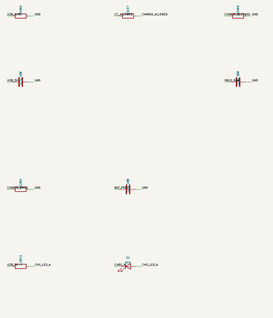
TP5670070 PG is low with EN low: dividers discharge through their lower resistors.

08 / 4.8 V SUPERCAP CHARGER + AUTOMATIC BALANCING



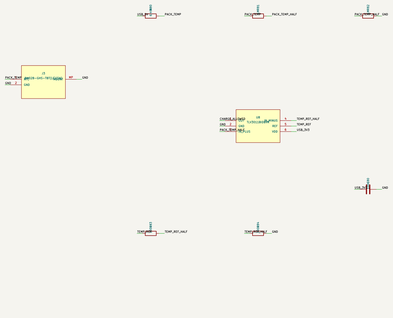
Charge enabled by ARM: USB presence or timed flash window forces SHDN low.

11 / PROTECTED VBUS + 300 mA CELL CHARGER



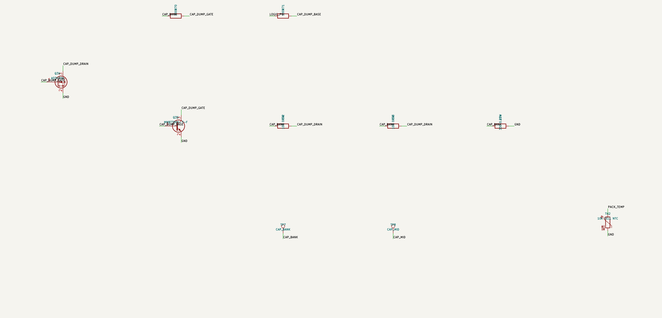
Charging disables logic and cap recharge in hardware. Ground is protected pack negative.

12 / CELL NTC + SHORTED-SENSOR INHIBIT



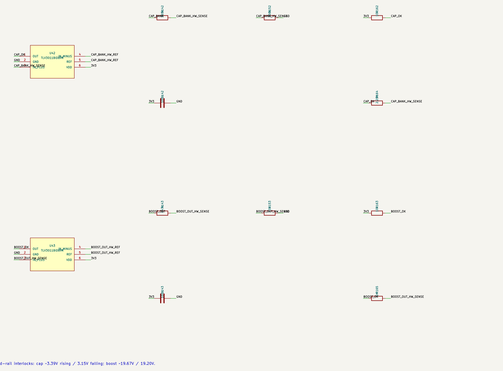
Open sensor inhibits through TP4056: short inhibits through U8. Check startup and thresholds.

13 / DEFAULT-ON CAPACITOR SERVICE DISCHARGE



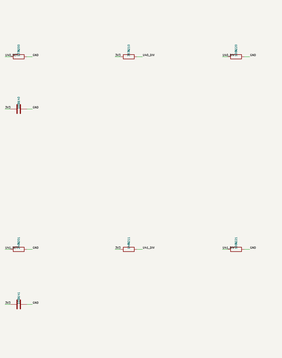
Active dump removes most energy; permanent 330ohm bleed provides final discharge. Measure TP7 before service.

14 / INDEPENDENT RAIL INTERLOCKS



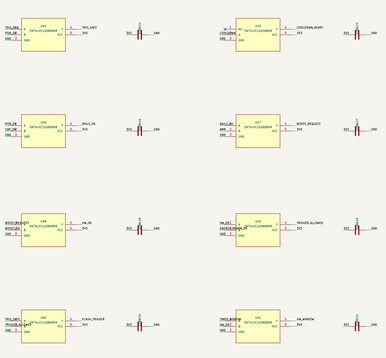
Loaded-rail interlocks: cap ~5.39V rising / 5.15V falling; boost ~19.47V / 19.20V.

15 / NON-RETRIGGERABLE PULSE + COOLDOWN

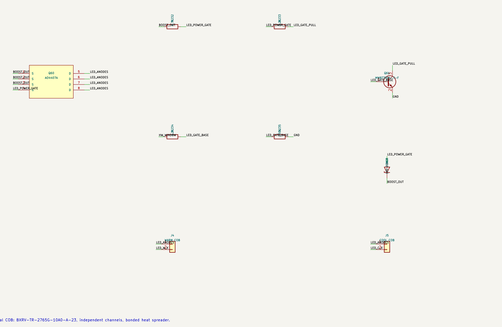


UVLO: 1.403ms nominal maximum window; UVLI: 1.258s lockout from each window start.

16 / HARDWARE FLASH PERMISSION + CHANNEL GATING

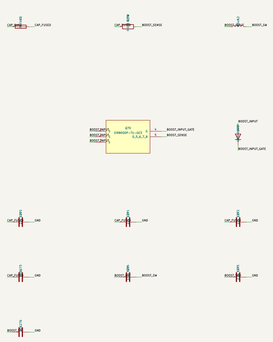


17 / PHYSICAL LED POWER CUTOFF — DEFAULT OFF



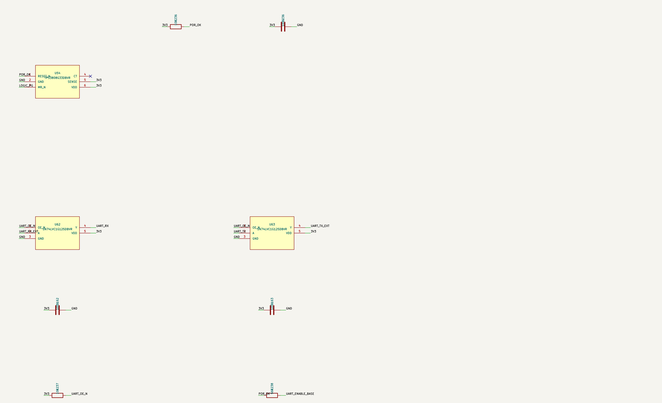
External COB: BXRV-78-2765G-1040-I-23, independent channels, bonded heat spreader.

18 / 21.07 V BOOST + INPUT DISCONNECT / CURRENT LIMIT

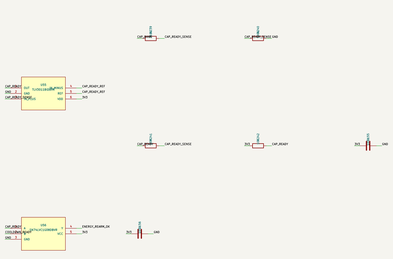


Output reservoir is BEFORE physical cutoff. Never add bulk capacitance on LED_ANODES.

19 / POWER-ON DELAY + Ioff UART ISOLATION



20 / PRE-FLASH ENERGY QUALIFICATION



Nominal cap-ready ~4.51V rising / 4.43V falling. Pulse sag is handled by UVLO.

Incomplete electrical design — NOT FOR FABRICATION

Sheet: /
File: round_LED_flash.kicad_sch

Title: ROUND LED FLASH — ENGINEERING DRAFT

Size: User	Date: 2026-05-20
KiCad E.D.A. 10.0.1	Rev: A-draft
	Id: 1/1